Interop 2018

Belgian Implementation of EIF-EIS-EIRA at Federal Level

2018-02-23

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DG Transformation digitale
SPF Stratégie et Appui

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Drivers & strategic goals

Why do we need a reference architecture?

Belgian Government:

The Belgian Agenda & Strategy

DG Digital Transformation Office's main functions:

✓ **Supporting** federal government with digitalization

✓ Driving force behind innovation & **digital transformation** of the federal government

✓ **Advizing** on technology & projects from an **enterprise and citizen centric perspective**

DG Digital Transformation Office's main services:

✓ Developing the digital **strategy and standards** for the federal government

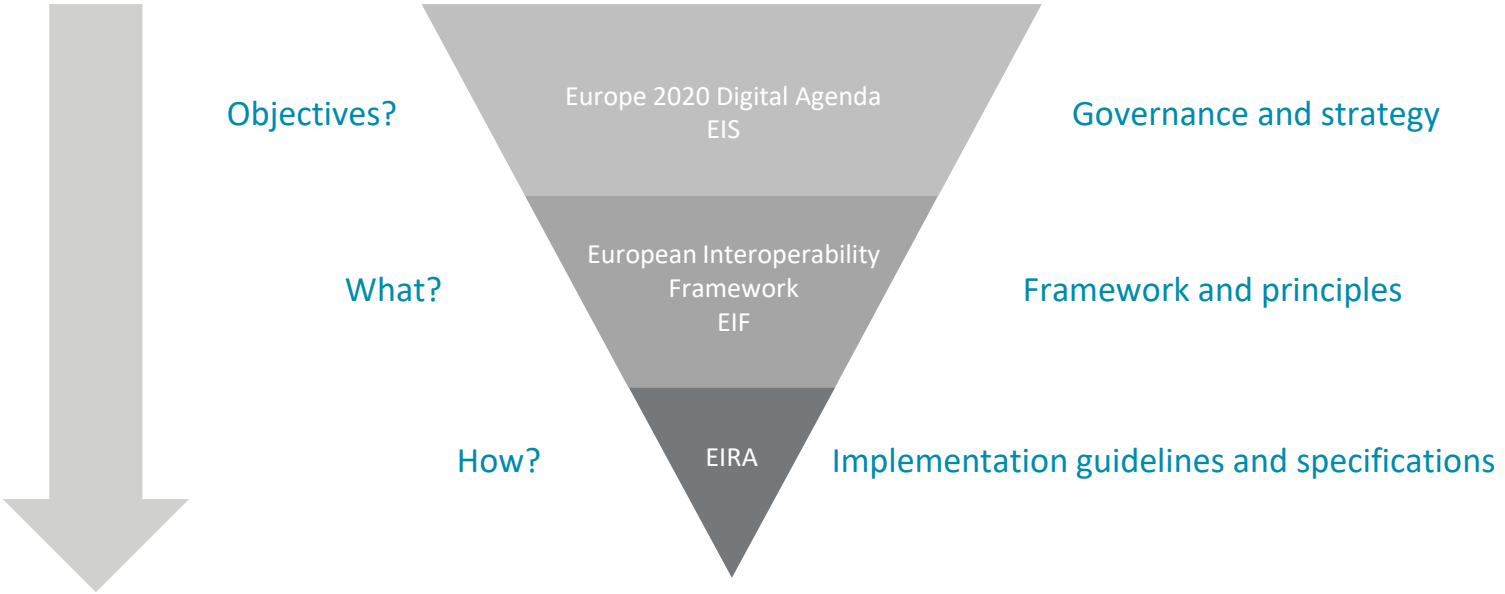
✓ Executing and **monitoring** the digital transformation of the federal government

✓ **Developing transversal digital services** and platforms like (Digital Communication services, Life events, Operations & Infrastructure Services,…)

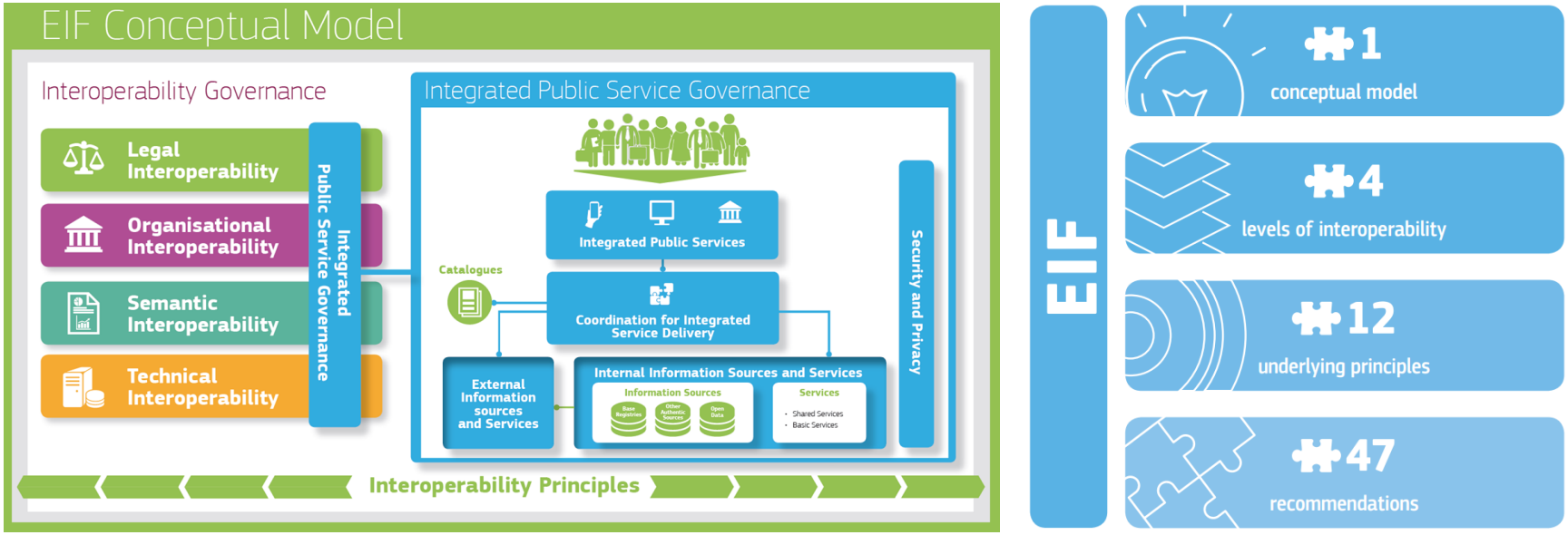
✓ **Overseeing the G-Cloud** strategic board and committees while reporting to the federal government

European Commission:

The European Digital 2020 Agenda & Strategy



European Interoperability Framework – EIF



"By following the EIF recommendations, EU countries currently digitizing their public administrations will follow a common approach when making their public services available online, integrate them end-to-end, manage their information sources, or deal with security and data protection rules.

*This way, public administrations can to **save time**, **reduce costs**, **increase transparency**, and **improve the quality of services** that they offer to citizens and businesses."*

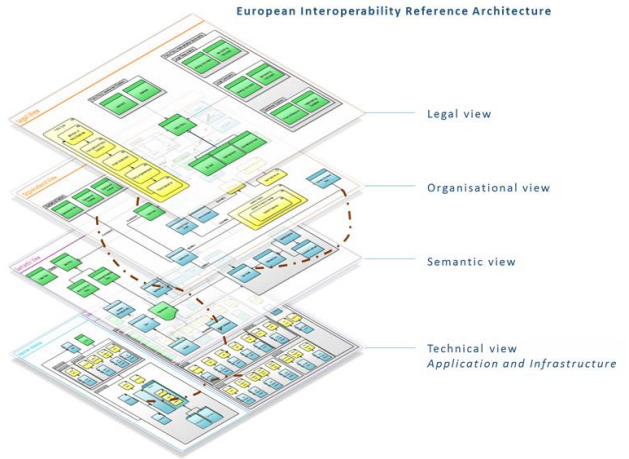
https://ec.europa.eu/isa2/eif_en

European Interoperability Reference Architecture – EIRA

- ✓ An architecture reference model
- ✓ Defines Architecture Building Blocks needed to build interoperable solutions supporting public services
- ✓ Based on the same 4 levels of interoperability described by the EIF
  - ✓ Legal view
  - ✓ Organisational view
  - ✓ Semantic view
  - ✓ Technical view

European Commission

EIRA
EUROPEAN INTEROPERABILITY REFERENCE ARCHITECTURE



European Interoperability Framework :

EIF -> BelgIF accepted by regions, language communities and at federal level

The overall report… (imho)

- Subsidiarity and proportionality 
- User centricity 
- Inclusion and accessibility 
- Security and privacy 
- Multilingualism 
- Administrative simplification 
- Transparency 
- Preservation of information 
- Openness 
- Reusability 
- Technological neutrality and adaptability 
- Effectiveness and efficiency 

EIF recommendations

Out of the principles, the framework distills 47 recommendations. These can serve as a guideline, or a checklist to determine the work a country should do to obtain or increase internal and external interoperability.



Recently, an update was published, with adapted recommendations that take into account new EU legislation (eIDAS, GDPR, PSI, Inspire, …)

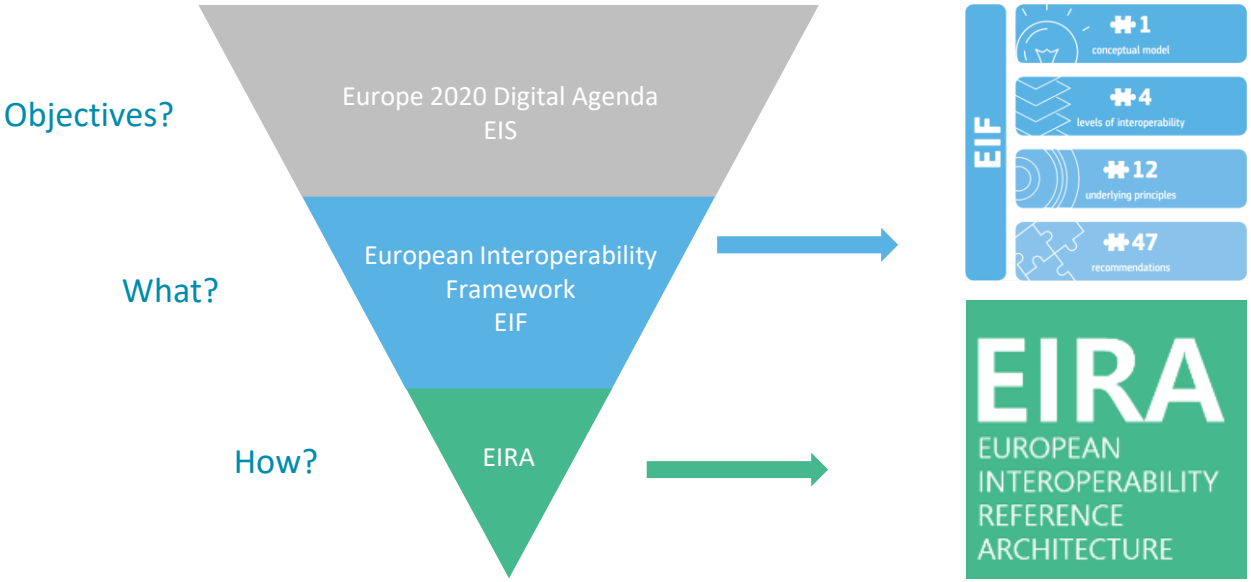


Belgium implements most of the recommendations
– partially in some cases, because of complexity and subsidiarity (silo thinking)



European Commission:

The European Digital 2020 Agenda & Strategy



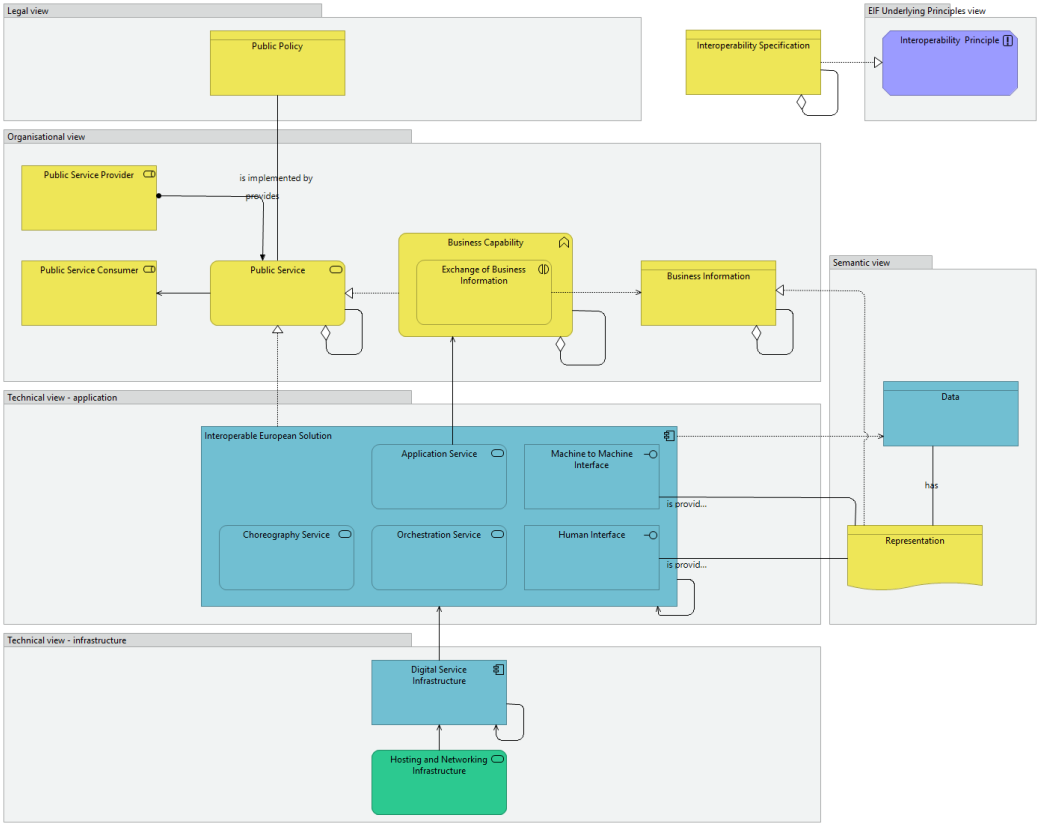
DTO Reference Architecture 2.0

A general overview

DTO Reference Architecture 2.0 – A general overview

Based on the same 4 levels of European Interoperability Framework

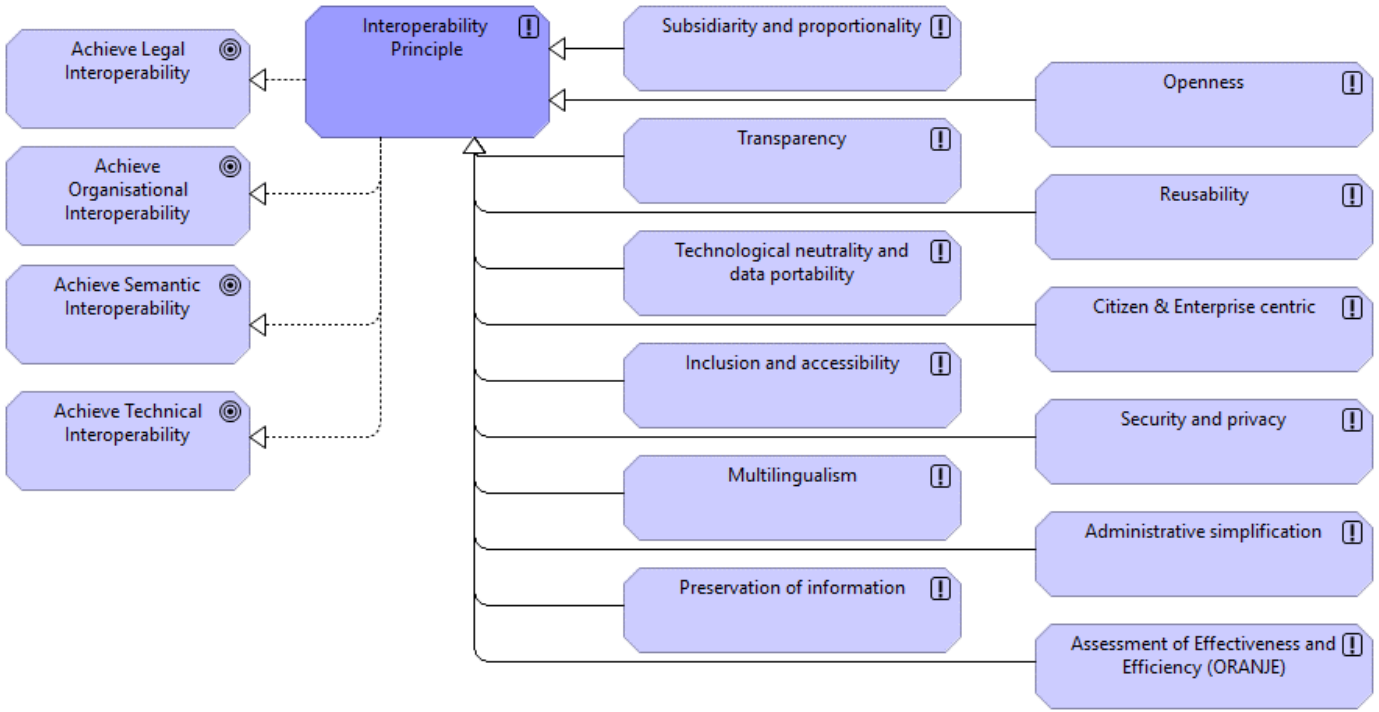




DTO Reference Architecture 2.0

The Motivational View

DTO Reference Architecture 2.0 – The Motivational View

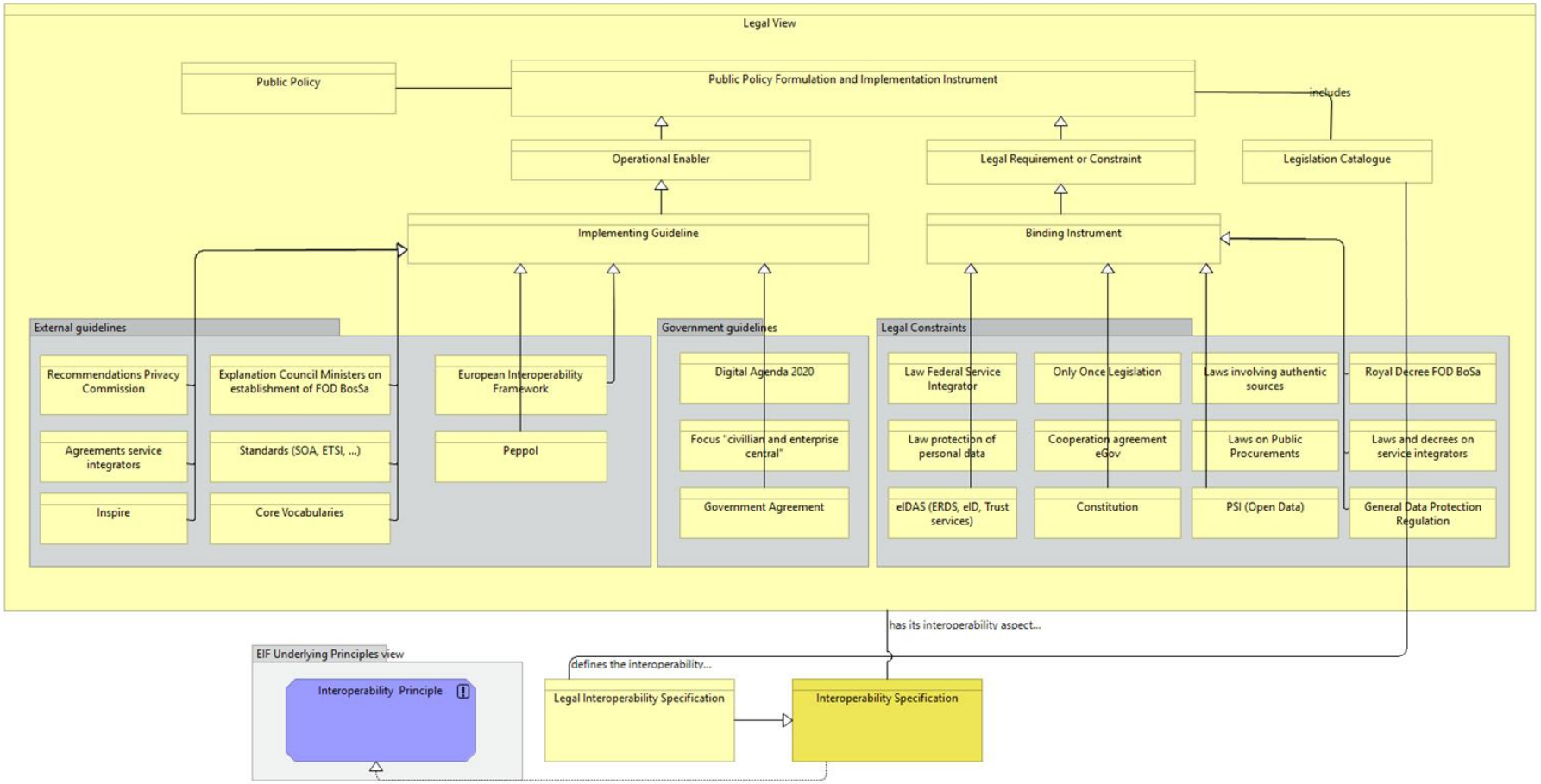


"The interoperability principles described by the EIF are fundamental behavioural aspects **to drive interoperability**. They describe the context in which public services are designed and implemented."

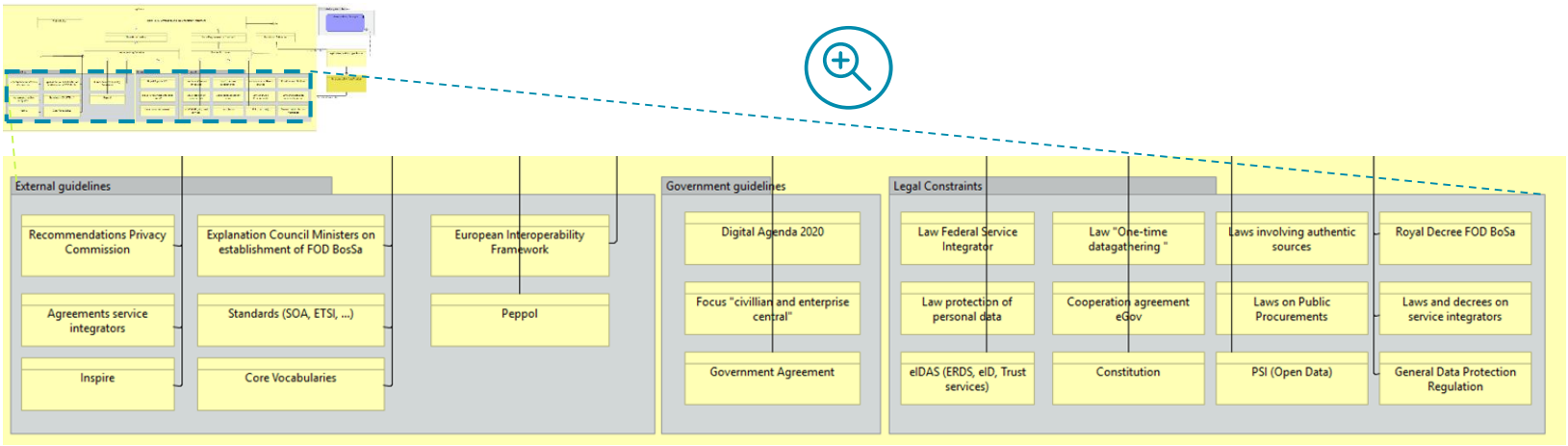
DTO Reference Architecture 2.0

The Legal View

DTO Reference Architecture 2.0 – The legal view



DTO Reference Architecture 2.0 – Legal View



External guidelines:

- ✓ Core Vocabularies
- ✓ Peppol
- ✓ …

Government guidelines:

- ✓ Digital Agenda 2020
- ✓ OSLO[2]

Legal Constraints:

- ✓ eIDAS
- ✓ General Data Protection Regulation (GDPR)
- ✓ …

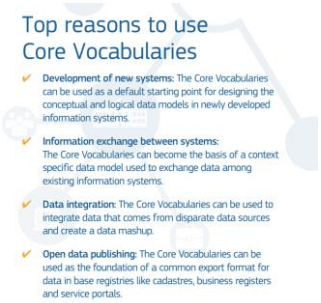
Implementing guidelines or general rules, principles, or pieces of advice to put a public policy into effect.

Binding instruments or legal means, involving an obligation, which are available to the European institutions to carry out their tasks.

DTO Reference Architecture 2.0 – Legal View

European Guidelines:

✓ EU Core Vocabularies



✓ PEPPOL

✓ INSPIRE

Belgian Governmental Guidelines:

✓ Digitale Agenda 2020

✓ Government Agreement – Regeerakkoord

✓ Federal Service Integrator Law – Wet op Dienstenintegratoren

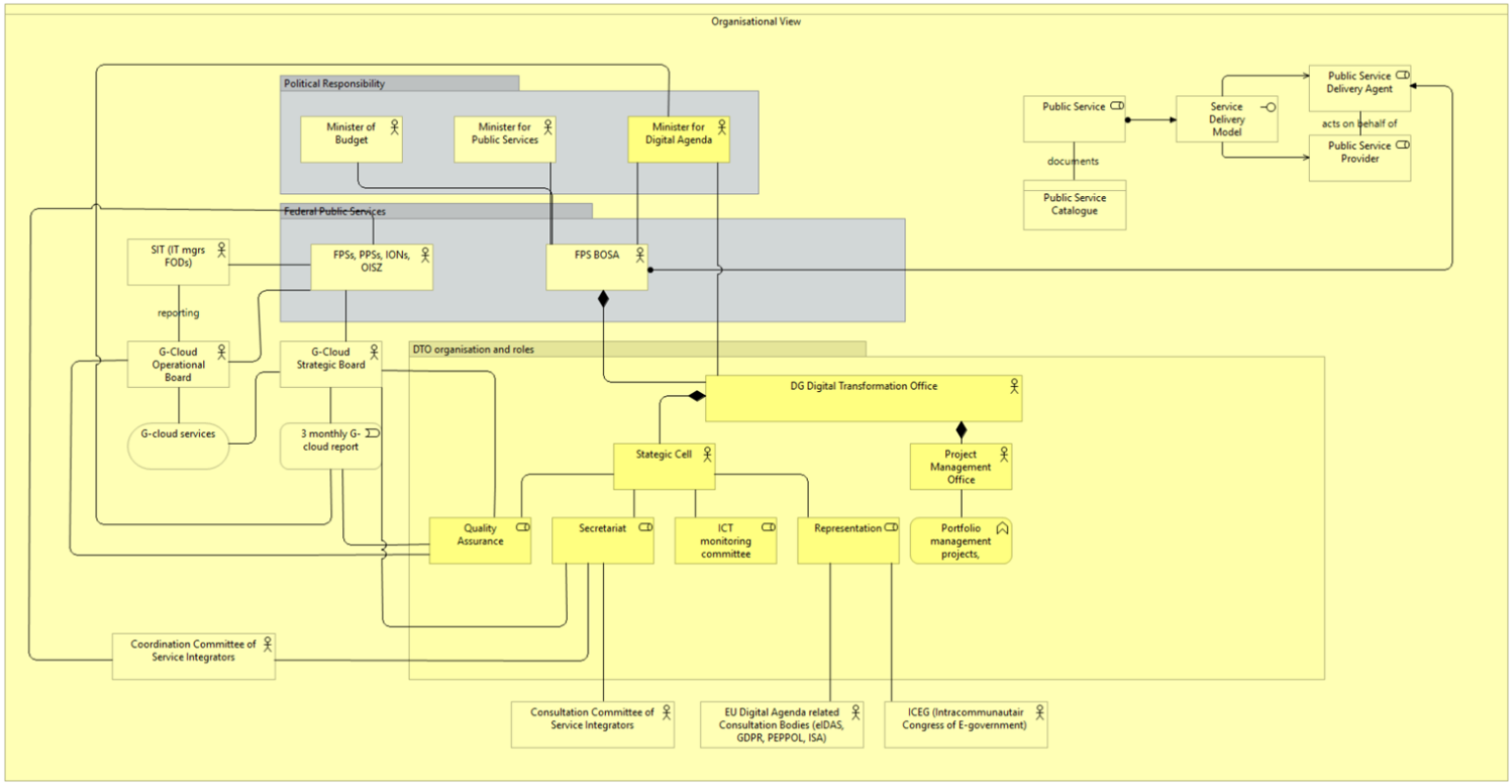
✓ OSLO^2 : Open Standaarden voor Linkende Overheden

Implementing guidelines or general rules, principles,
or pieces of advice to put a public policy into effect.

DTO Reference Architecture 2.0

The Organisational View

DTO Reference Architecture 2.0 – The Organisational View

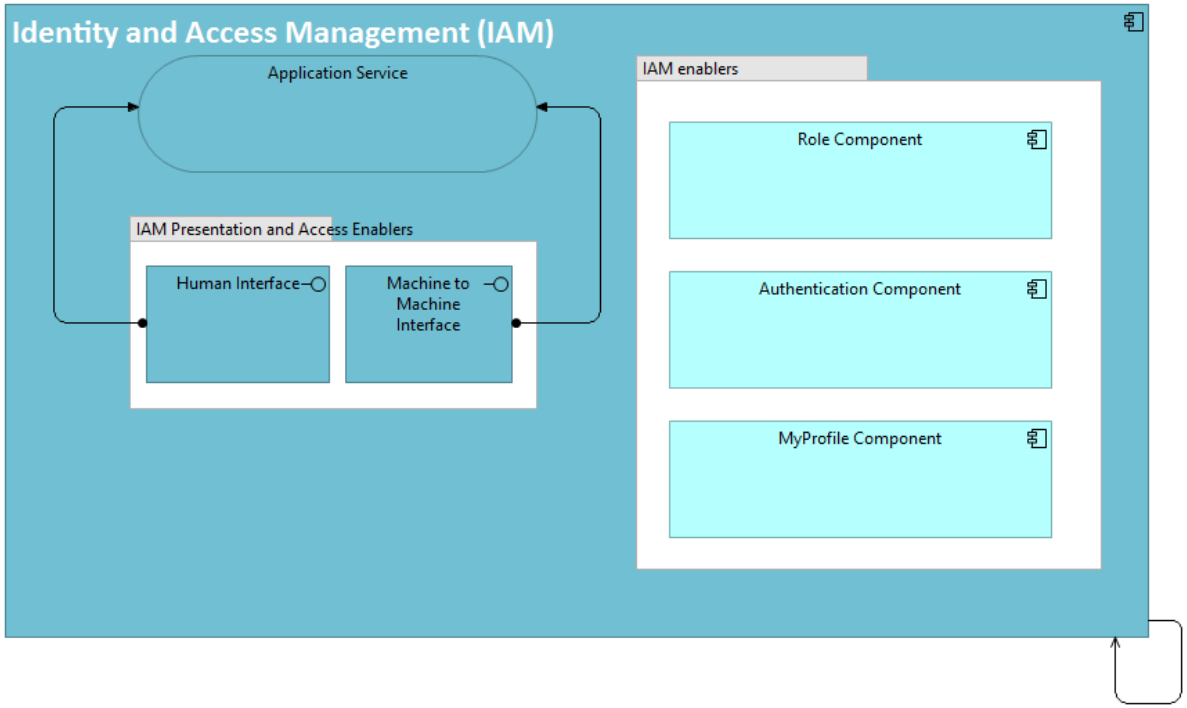


DTO Reference Architecture 2.0

The Application View

DTO Reference Architecture 2.0 – Application View

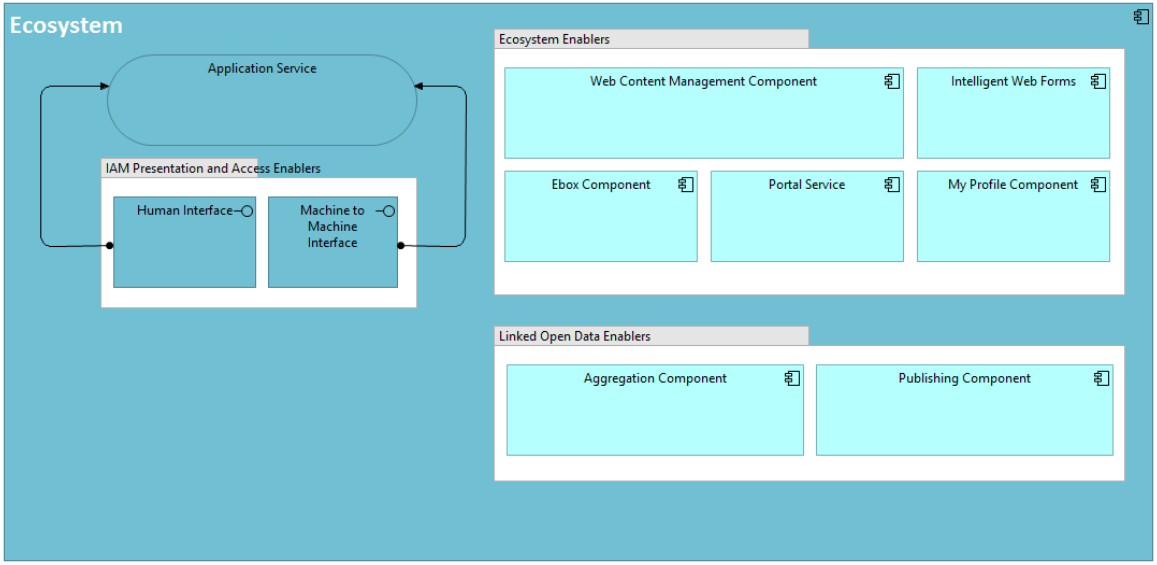
IAA



- ✓ Role Component:
  This is where roles & mandates are assigned, defined and managed.
- ✓ Authentication Component:
  Consist of an Identification, Authentication and Customer Management Service
- ✓ MyProfile Component:
  Consists of My Digital Keys and My Info. Services still in development phase are My Consent, My Preferences & My Security Settings

DTO Reference Architecture 2.0 – Application View

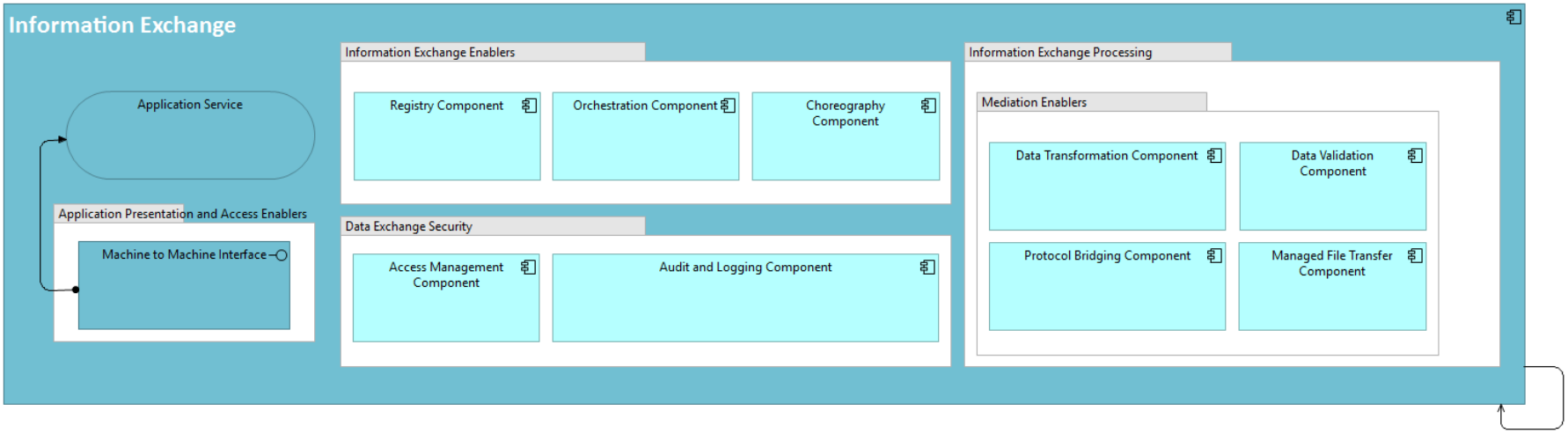
Ecosystem



- ✓ Web Content Management Component:
  Supports 200+ websites through the latest Drupal content management platform.
- ✓ Intelligent Web Forms Component:
  Promotes and facilitates reuse and interoperability of input data through the application of the “Only Once” principle. Input information by citizen/service should only be asked once & be reused.
- ✓ Ebox Component:
  Electronic mailbox for official documents, tax receipts, Zoomit & Doccle integration,…
- ✓ Linked Open Data Enablers:
  Facilitates public and private open data sets.

DTO Reference Architecture 2.0 – Application View

Information Exchange



✓ Data Exchange Security:

These components provide the mechanisms to ensure that only authorized users can access and perform actions on IT resources as well as providing support for accountability, auditing, tracing and logging actions.

✓ Information Exchange Processing:

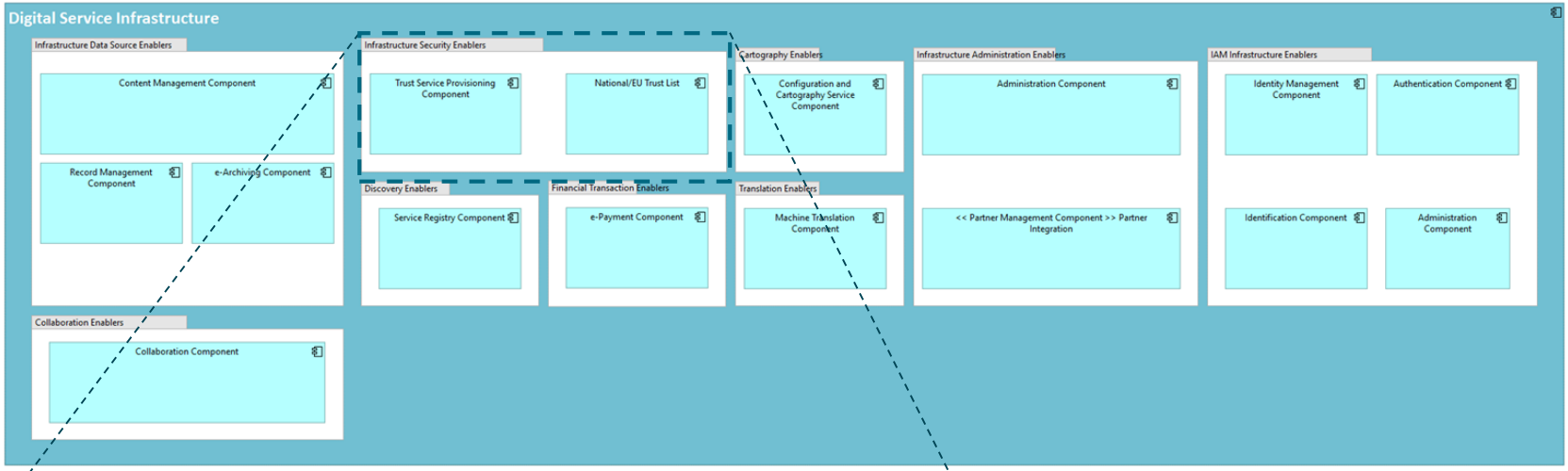
These components enable the transformation, conversion, validation of data and technical translation between protocols

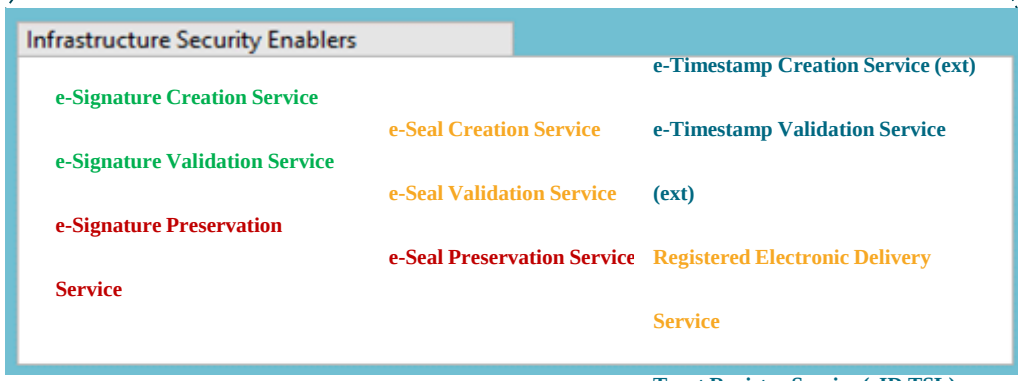
✓ Information Exchange Enablers:

The Registry component consists of a registry of all governmental web services whereas the Orchestration and Choreography components facilitate the exchange of information between them.

DTO Reference Architecture 2.0 – Infrastructure View

Digital Service Infrastructure





✓ Infrastructure Security Enablers:
The underlying components enable the creation, verification, validation and preservation of electronic signatures, seals, time stamps and certificates.

Thank you !

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